UNIVERSITY OF CENTRAL PUNJAB LAHORE
Faculty of IT & CS
Mid-term Examination

Degree Program	BSCS	Semester	Fall 2023
Course Title	Multivariate Calculus	Course Code	CSSS2733
Course Coordinator	Muhammad Rayees Ahmad	Total Marks	60
Total time	90 minutes	Section	
Student Name		Registration #	

Instructions:

- Write your name and registration number on the question paper and answer sheet in the provided spaces.
- Write with blue/black pen/pointer/ballpoint only. Do not use any other colored pen.
- Finish your exam within the prescribed time. No extra time shall be given.
- Do not write direct answers. Marks will be given for correct procedure and complete solution.

Attempt all question.

Q # 1 (4+4+4=12 marks): Consider the points $R\ (3,-1,0)$, $S\ (-1,0,4)$ and $T\ (0,6,-7)$. then

a) <u>Find</u> the vectors $\vec{a} = \overrightarrow{RS}$ and $\vec{b} = \overrightarrow{RT}$.

b) <u>Find</u> $\vec{a} \times \frac{1}{2}\vec{b}$

c) <u>Find</u> the angle between $\vec{a}$ and $\vec{b}$. (Minimum)

Q # 2 (5+7=12 marks):

(a) <u>Find</u> the equation of plane passing through the point $(1,-1,0)$ and parallel to the plane $x - 3y + 2z = 0$.

(b) <u>Find</u> the distance of a point $(8,-10,-5)$ from the plane $-3x + 2y + 4z - 11 = 7$.

$x = x_0 + vt,$
$y = y_0 + vt$
$z = z_0 + vt$

Q # 3 (10+10=20 marks):

(a) <u>Determine</u> the **parametric equations** for the line of intersection of the two planes given below:

$Plane\ 1\colon\ x + 3y + 2z - 6 = 0$ $\qquad$ $Plane\ 2\colon\ 2x - y + z + 2 = 0$

(b) <u>Examine</u> whether the given lines are **parallel, perpendicular, intersecting, or skew lines.** If the lines are intersecting lines, then <u>find</u> the **point of intersection**.

$Line\ L_1\colon \quad x = 2t + 1; \quad y = 3t + 2; \quad z = 4t + 3$

$Line\ L_2\colon \quad x = 5u + 4; \quad y = 2u + 1; \quad z = u$

Q # 4 (4+4+4+4=16 marks):

Consider the **velocity vector function**

$$\vec{v}(t) = \sin(4t)\,\hat{\imath} + \left(\frac{t^2 - 1}{\sqrt{7}}\right)\hat{\jmath} + \left(2 - e^{-3t}\right)\hat{k}$$

a) <u>Find</u> the **domain** of the velocity vector function $\vec{v}(t)$.

b) <u>Find</u> the **displacement** vector function $\vec{r}(t)$.

c) <u>Find</u> the **distance** at $t = 1$, when $c = 0$.

d) <u>Find</u> the **acceleration** vector function $\vec{a}(t)$ at $t = 1$.

(Note: Round off your answer up to 2 decimal places).

$L_1 \times L_2$

$= KV.$